INCH-POUND

FF-S-111E
1 MARCH 2022
SUPERSEDING
FF-S-111D
27 AUGUST 1974

DETAIL SPECIFICATION SHEET

SCREW, WOOD

Inactive for new design after 04 December 2000
and is no longer used, except for replacement purposes.

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

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AMSC N/A FSC 5305

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1. SCOPE

1.1. Scope. This specification covers single and double-threaded wood screws with round, flat countersunk and oval countersunk heads having slotted or crossed recessed drives.

1.2. Classification.

1.2.1. Type and style. Wood screws shall be of the following types and styles (see 6.1)

Type:
- I - Single thread
- II - Double thread

Style:
- 1s - Round head, slotted drive.
- 1c - Round head, cross-recessed drive.
- 2s - Flat countersunk head, slotted drive.
- 2c - Flat countersunk head, cross-recessed drive.
- 6s - Oval countersunk head, slotted drive.
- 6c - Oval countersunk head, crosse-recessed drive.

1.2.2. Size. Wood screws shall be of the sizes specified in the applicable Military Standard (MS) and ASME B18.6.1.

2. APPLICABLE DOCUMENTS

2.1. General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2. Government Documents.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916 – DOD Preferred Methods for Acceptance of Product.

MIL-PRF-32647 – ZINC-NICKEL ELECTROPLATING FOR FASTENERS

(Copies of this documents are available online at https://quicksearch.dla.mil.)

2.3. Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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AMERICAN SOCIETY FOR QUALITY (ASQ).

ASQ H1331 – Zero Acceptance Sampling Plans

(Copies of this document are available for purchase online from www.asq.org.)

THE AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME).

ASME B18.6.1 – Wood Screws

(Copies of this document are available for purchase online from www.asme.org.)

ASTM INTERNATIONAL (ASTM).

ASTM A108 – Steel Bars, Carbon, Cold Finished Standard Quality

ASTM A576 – Standard Specification for Steel Bars, Carbon, Hot-Wrought, Special Quality

ASTM B633 – Zinc on Iron and Steel, Electrodeposited Coatings of

ASTM B36/B36M – Standard Specification for Brass Plate, Sheet, Strip, And Rolled Bar

ASTM A675/A675M – Standard Specification for Steel Bars, Carbon, Hot-Wrought, Special Quality, Mechanical Properties

ASTM B134/B134M – Standard Specification for Brass Wire

ASTM B98/B98M – Standard Specification for Copper-Silicon Alloy Rod, Bar and Shapes

ASTM B99/B99M – Standard Specification for Copper-Silicon Alloy Wire for General Applications

(Copies of this document are available for purchase online from www.astm.org.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE) INTERNATIONAL

SAE-AMS2700 – Passivation Treatments for Corrosion-Resisting Steel.

SAE-AMS-QQ-P-416 – Plating, Cadmium (Electrodeposited).

SAE-AMS-C-81562 – Coatings, Cadmium, Tin-Cadmium and Zinc (Mechanically Deposited).

SAE AIR4127 – Steel: Chemical Composition and Hardenability

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2.4. Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained."

3. REQUIERMENTS

3.1. Material. Wood screws shall be made from the following materials, as specified (see 6.1).

3.1.1. Carbon steel. Carbon steel shall be grades 1010 (UNS G10100) thru 1022 (UNS G10220) or Grades 1108 (UNS G11080) thru 1118 (UNS G11180) in accordance with ASTM A108, ASTM A576 or ASTM A675/A675M as applicable. Minimum ultimate tensile strength shall be 55,000 PSI.

3.1.2. Corrosion-resisting steel. Corrosion-resisting steel shall be types 302 (UNS S30200), 303 (UNS S30300), 304 (UNS S30400), 305 (UNS S30500), 316 (UNS S31600) or 384 (UNS S38400) in accordance with SAE AIR4127 as applicable. Minimum ultimate tensile strength shall be 70,000 PSI.

3.1.3. Brass. Brass shall be alloy no. 260 (UNS C26000) or 268 (UNS C26800), quarter-hard in accordance with ASTM B36/B36M; or alloy no. 260 (UNS C26000) or 270 (UNS C27000), quarter hard in accordance with ASTM B134/B134M. Minimum ultimate tensile strength shall be 50,000PSI.

3.1.4. Copper-silicone alloy. Copper-silicone alloy shall be alloy no. 651 (UNS C65100) in accordance with ASTM B98/B98M or ASTM B99/B99M as applicable, minimum ultimate tensile strength shall be 50,000 PSI.

3.2. Protective finish. Wood screws shall be furnished plain (uncoated), with a naturally bright finish, except as noted in 3.2.1 thru 3.2.4.

3.2.1. Galvanizing. Wood screws for pole-line use shall be galvanized in accordance with SAE-AMS2700 method 1.

3.2.2. Passivation. Corrosion-resisting steel wood screws shall be passivated in accordance with SAE-AMS2700 Method 1.

3.2.3. Cadmium Plating. When specified (see 6.1), carbon steel wood screws shall be cadmium plated in accordance with Type II, Class 3 of SAE-AMS-QQ-P-416 or SAE-AMS-C-81562.

/1 Cadmium is an R1, R2, and D1 substance in the Aerospace and Defense Declarable Substance List. Zinc-Nickel plate per MIL-PRF-32647, Type K16 is the preferred method.

3.2.4. Zinc-Nickle Plate. When specified (see 6.1), carbon steel wood screws may be Zinc-Nickel plated as an alternative to cadmium plate in accordance with MIL-PRF-32647, Type K16.

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3.2.5. Zinc coating. When specified (see 6.1), carbon steel wood screws shall be electrodeposited zinc coated in accordance with ASTM B633, Fe/Zn 5, Type II or mechanically deposited zinc coated in accordance with SAE-AMS-C-81562, Type II, Class 3.

3.3. Dimensions.

3.3.1. Type I screws. Dimensions of type I, single-thread screws shall be in accordance with the applicable Military Standard (MS) and ASME B18.6.1.

3.3.2. Type II screws. Thread and shank diameters and threads per inch for type II, double-threaded screws shall be in accordance with Table I. Other dimensions shall be in accordance with ASME B18.6.1.

TABLE I. Dimensions (in inches) of type II double-thread screws

Nominal Size	Threads Per Inch	Outside Diameter of Threads +.004 -.007	Diameter of unthreaded shank +.010 -.000
2	28	.086	.066
3	26	.099	.081
4	24	.112	.090
5	22	.125	.102
6	20	.138	.111
7	18	.151	.122
8	18	.164	.131
9	16	.177	.141
10	15	.190	.152
12	13	.216	.171
14	12	.242	.196
16	10	.268	.209
18	10	.294	.232

3.4. Workmanship. Screws shall be free from fins, burrs, sharp edges, cracks, and surface contamination.

4. QUALITY ASSURANCE PROVISIONS

4.1. Responsibility for inspection. Unless otherwise specified in the contractor purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the

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inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2. Lot. A lot shall consist of all screws of the same material, type, style, size, and protective finish, produced under the same conditions and offered for acceptance at one time.

4.3. Sampling for examination. Unless otherwise specified in thew contract (see 6.1), for sampling for examination, an “accept at zero” (c=0) sampling plan such as MIL-STD-1916 or ASQ H1331 shall apply. Acceptable verification levels for MIL-STD-1916 and index values for ASQ H1331 shall be specified in Table II.

4.4. Examination. Each screw taken as specified in 4.3 shall be examined to verify conformance with this specification. Examination shall be conducted in accordance with Table II. Any screw in the sample containing one or more defects shall be rejected and if the number of defective screws in any category exceeds the acceptance number for that category, the lot represented by the sample shall be rejected.

Table II. Classification of Defects.

Category	Defect	Inspection Method
Critical	None defined	
Major	Verification Level II, Index Value = 4.0	
101	Size (1.2.2)	SIE*
102	Length (3.3)	SIE
103	Slot recessed dimensions (3.3)	SIE
104	Thread (3.3)	SIE
Minor	Verification level I, Index Value = 6.5	
201	Other dimensions (3.3)	SIE
202	Protective finish, missing or incomplete (3.2)	Visual
203	Workmanship (3.4)	Visual

*SIE=standard Inspection Equipment

4.5. Physical test. Normally a physical test will be required. When physical tests are required, they shall be conducted in accordance with the method specified (6.1).

4.6. Protective finish. Examination and test of protective finishes shall be in accordance with the specification of 3.2.

5. PACKAGING

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5.1. Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.1). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1. Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

a. Title, number, and date of the specification.
b. Type, style, and size (1.2).
c. Material (3.1).
d. Protection finish (3.2).
e. Tests, if required (4.5).
f. Selection of applicable level of packaging and packing (5.1).
g. Applicable MS part number.
h. Marking, if required (5.1)

6.2. Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue."

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